



Weekly Report

Agenda

- Data Capture
- CFAR Progress
- Moving Forward



Data Capture

- 2 Data Captures at Lake Murray Spot
- Good Data?
- Good Pictures?
- Next Spot?

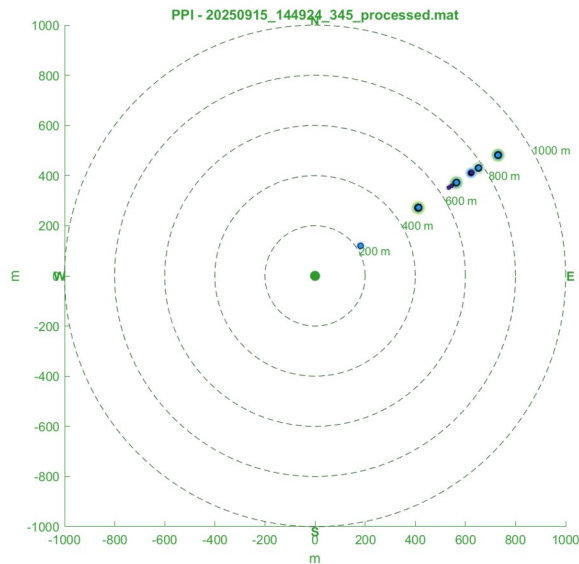


CFAR Results

- Using MATLAB/Python to create cleaned CFAR data
- Initial Troubles, still working out parameters

Generated Images from altered .csv files

Threshold in the negatives?

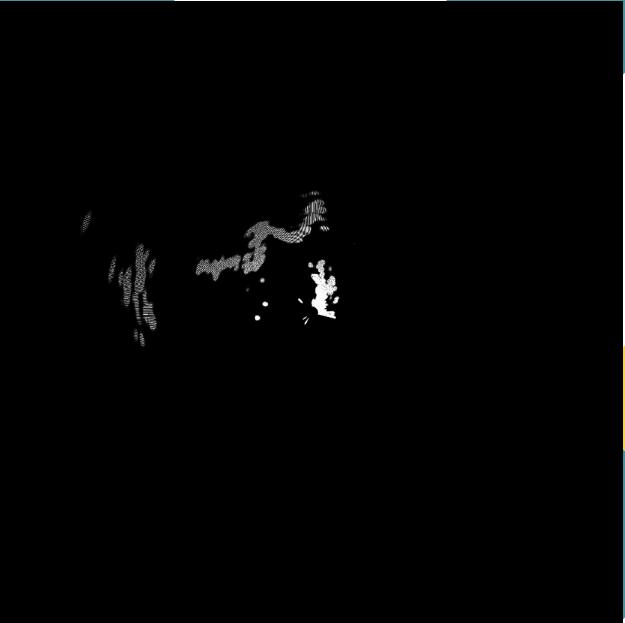


CFAR Detections (20250915_144924_345.csv)

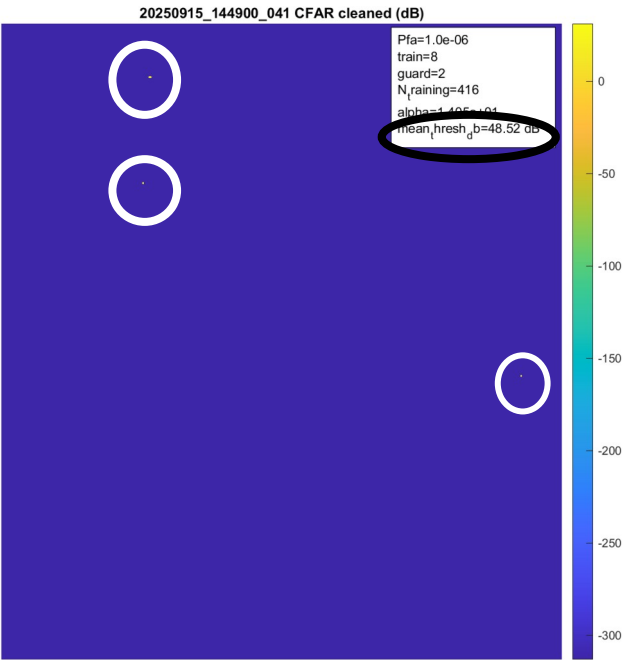
Pfa=1e-06
train=8
guard=2
N_training=416
alpha=1.405e+01
mean_thresh_db=-240.00 dB



CFAR Results Continued

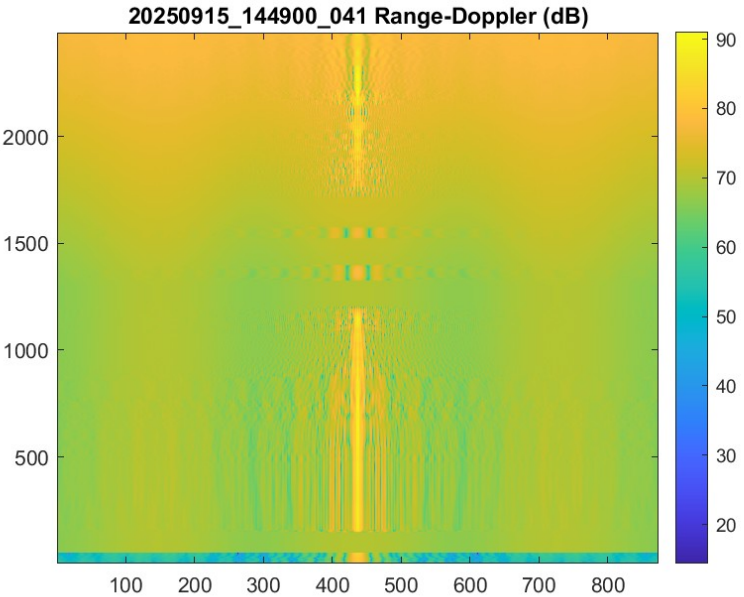


<- Actual Radar Image

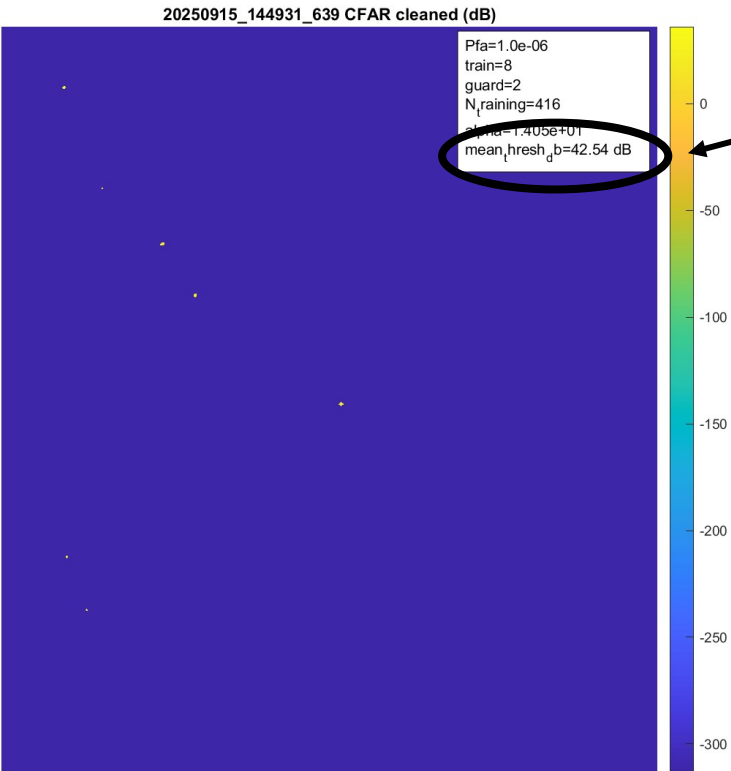


<- Cleaned Data
processed through
CFAR

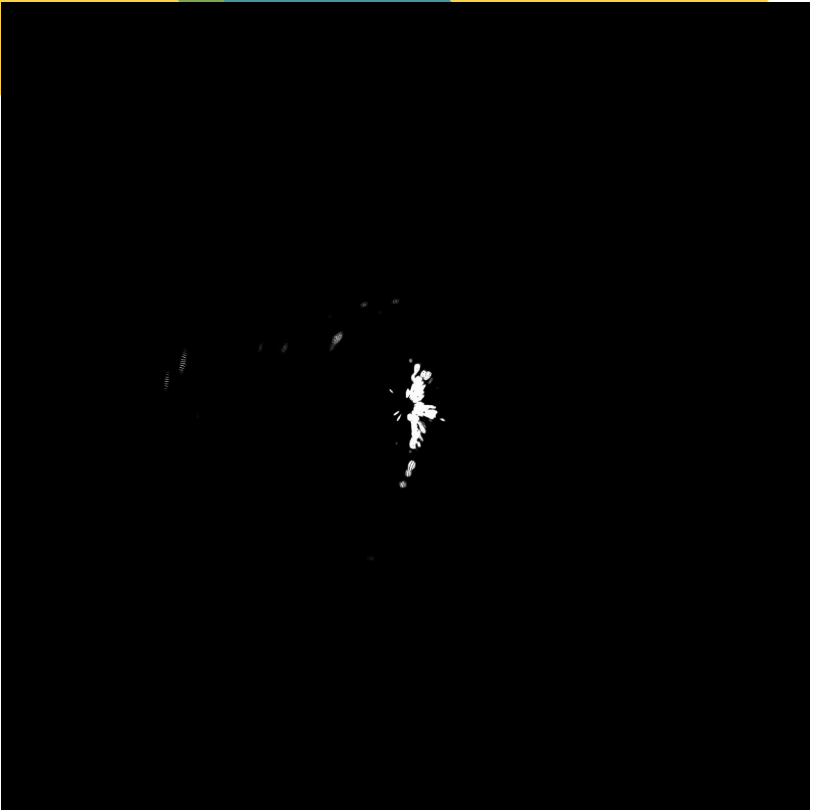
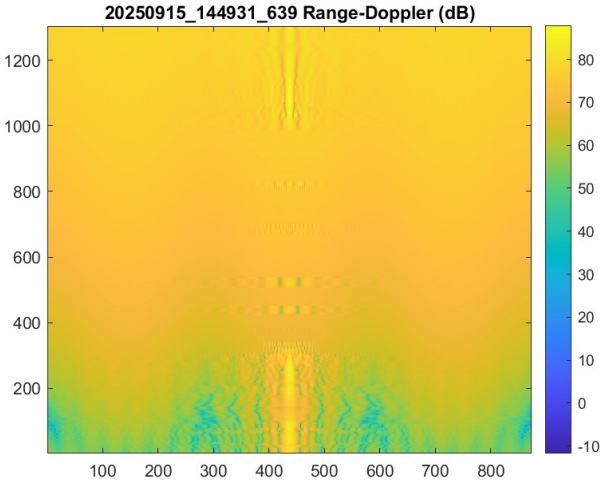
Raw Doppler Data ->



More CFAR Results



Change in
threshold allows
for more positive
target ID



My Interpretations

- The data I am collecting isn't noisy enough to make any targets highlight through the CFAR threshold -> create a pattern that picks up more noise
- The CFAR parameters need to be tweaked in order to get more precise decluttered images



Moving Forward

- Continue with weekly data capture at new locations, further distance location this weekend
- ~~Convert .csv files to .png radar images~~
- Test CFAR algorithms until an acceptable level of “Clean Data” is reached (???)
- Use simple data augmentation to make more radar data to train AI model

